

Society for Computer Technology & Research's
PUNE INSTITUTE OF COMPUTER TECHNOLOGY, Pune.
DEPARTMENT OF INFORMATION TECHNOLOGY

DSA Mini Project
ONE-PAGE GROUP REPORT

Class: SY - IT Division: E-11 Group No.: 5
Day & Date of Presentation: Tuesday, 03/11/26

Group members Name and Roll No.:

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Topic/Case Study Title: DNA Sequence Analyzer

Problem statement: DNA sequences contain mutations leading to genetic variations. Manually comparing them is time-consuming and error-prone. This project automates the comparison of verified original and mutated FASTA sequences to identify changes efficiently.

Objectives:

- 1. Calculate sequence similarity and Hamming distance between original and mutated DNA strings.
- 2. Identify and export the exact positions of nucleotide substitutions to a structured CSV report.

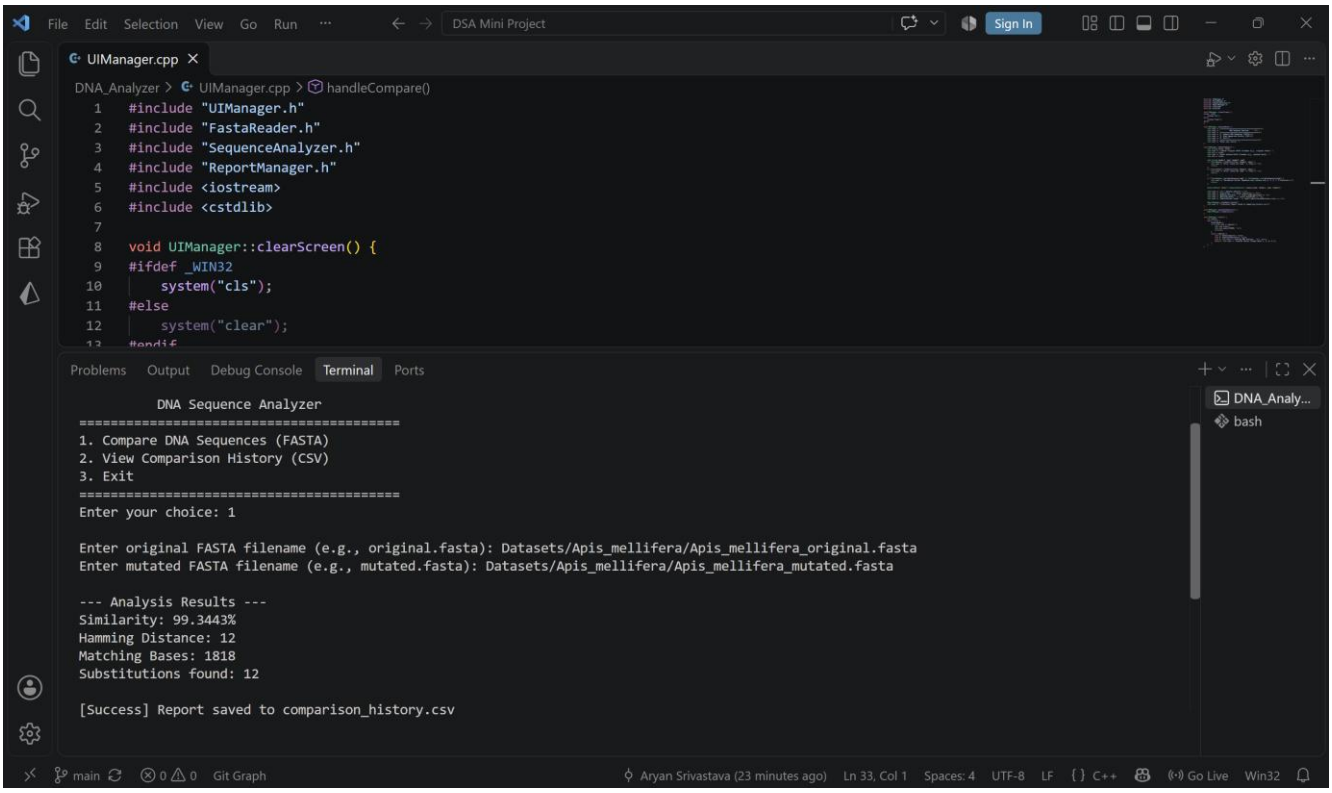
Data structures used (Data structure(s) and 1–2 line justification for each):

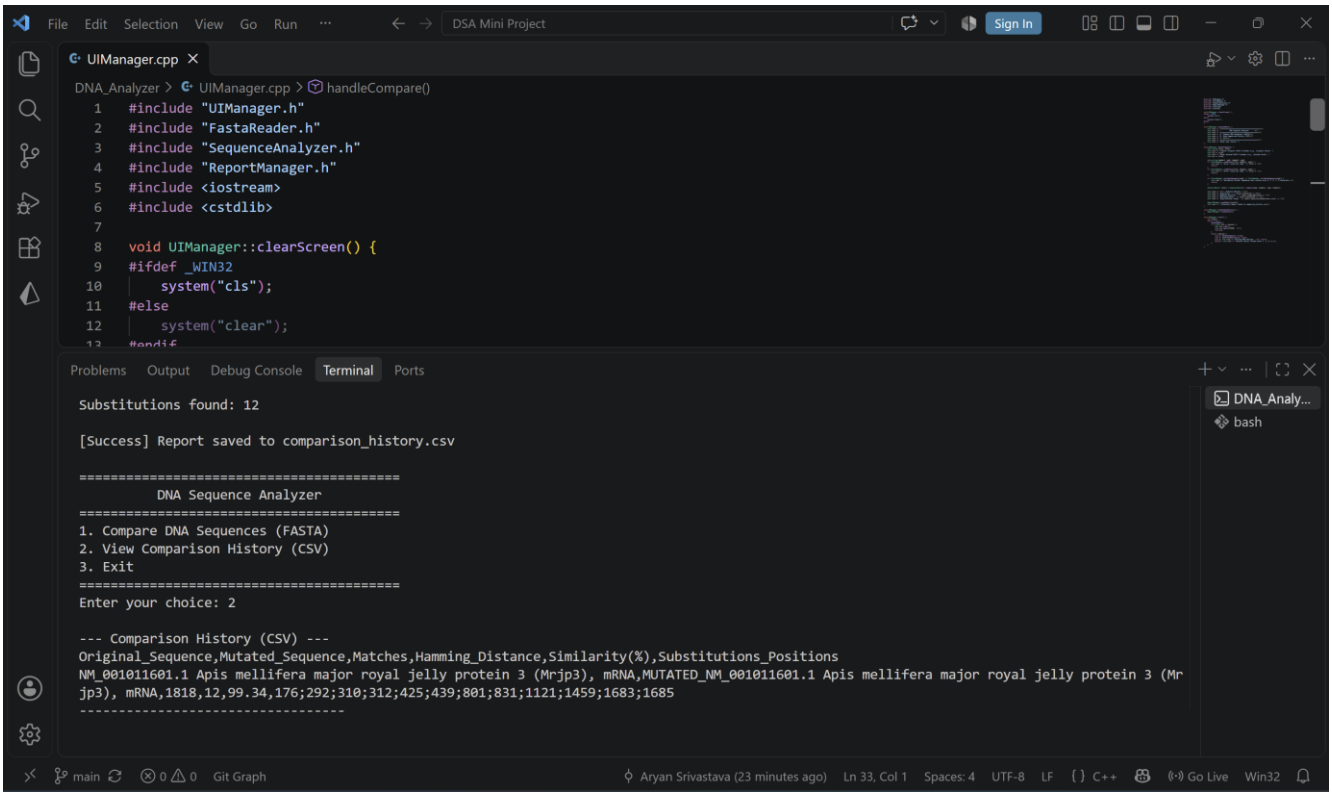
• Strings (std::string): Used to store FASTA headers and DNA sequences, allowing efficient O(1) character access and straightforward string comparisons.

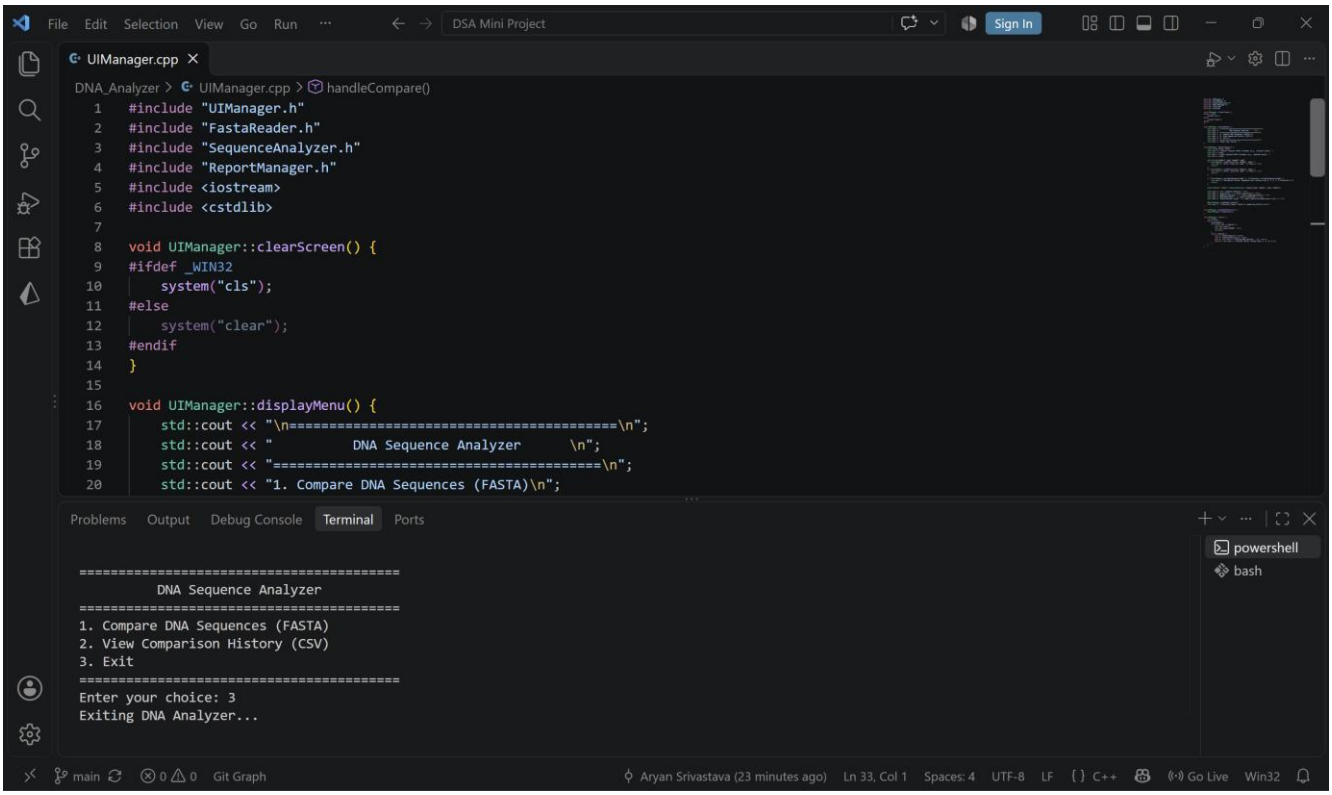
• Dynamic Arrays (std::vector<int>): Used to dynamically store the indices of nucleotide substitutions, as the exact number of mutations is unknown before runtime.

Implementation (Programming language / tools): C++17 (Standard Template Library), CMake

Implementation output:







Conclusion / learning Key outcome or learning (1–2 sentences): Successfully developed an automated tool for genetic sequence comparison that significantly reduces manual effort. Gained practical experience applying string manipulation and dynamic arrays to biological data processing.

Student signatures:

1. _____

2. _____

3. _____

4. _____

Faculty remarks: _____